Laboratory for Scientific Computing
UNIVERSITY OF CAMBRIDGE | Cavendish Laboratory
Department of Physics

# Mathematical Methods for Scientific Computing

## *Partial Differential Equations*

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# Partial differential equations

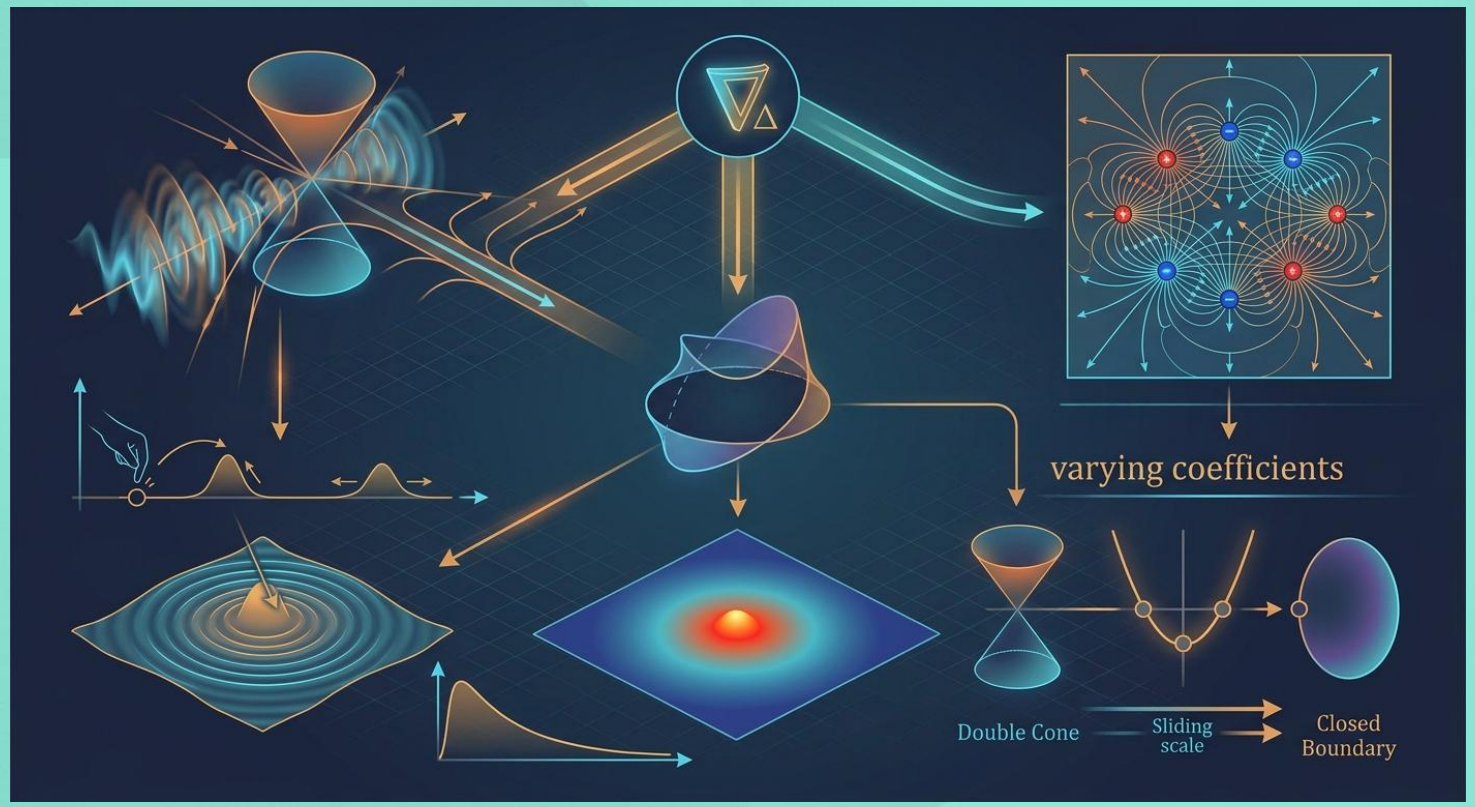
- The overall field of partial differential equations is large – you have likely covered many different techniques for solving these equations, e.g. separation of variables
- We are not going to cover that here – if we already have an analytic solution technique for the PDE, there is very little need to use scientific computing
- Instead, we are going to focus on how PDEs are used in scientific computing, which relates more to equation manipulation and classification, than solution
- A wide range of physical problems can be expressed as the solutions to PDEs, often very complex systems of PDEs
- In these cases, we would expect our computational methods to do the actual solving
- The challenge might often be expressing our system in a manner that can be solved - we need to know something about the behaviour of our system

# Outline

- 1. Classification of PDEs
- 2. Different forms for PDEs
- 3. Discontinuities and PDEs

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# *Classification of PDEs*

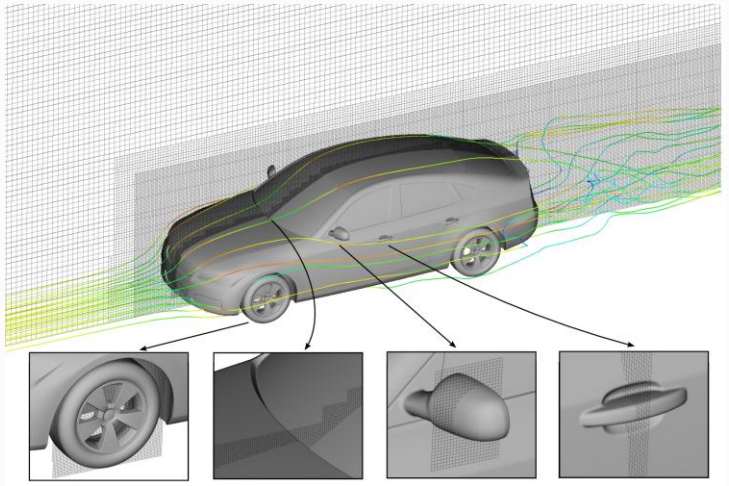


# Classification of PDEs

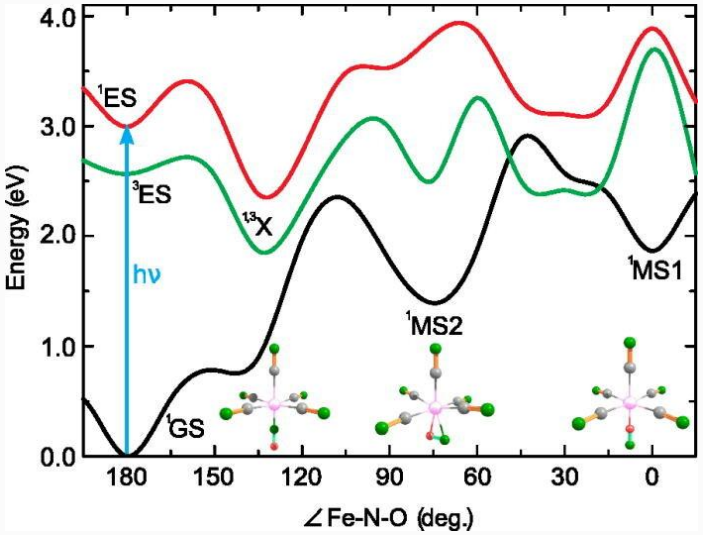
- There is no single technique for solving a PDE, or system of PDEs, neither analytically nor in scientific computing
- This is based on the mathematical and physical behaviour of the underlying problem
- By **classifying** our PDEs (a mathematical approach), we can attempt to identify whether they are capable of producing the correct physics we wish to capture
- And we can also ensure we can identify a computational approach which will allow this behaviour to be captured
- More accurately, we can identify a computational approach that is both suitably accurate and suitably efficient for the problem

# Steady state problems

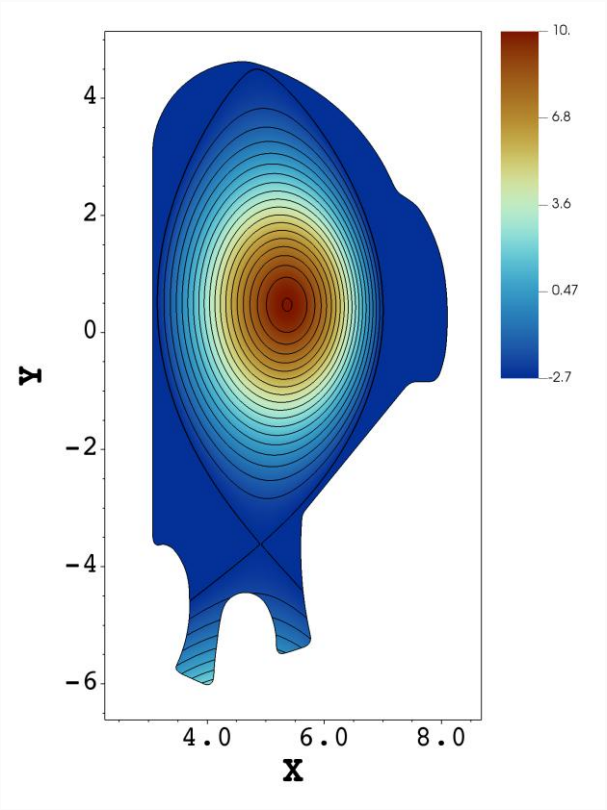
- One class of problem could be those where the solution is a fixed value (typically in time)
- This could be considered to be a balance of physical states, for example



- Airflow around an object, e.g. a car or aerofoil – from steady state conditions, lift and drag can be computed



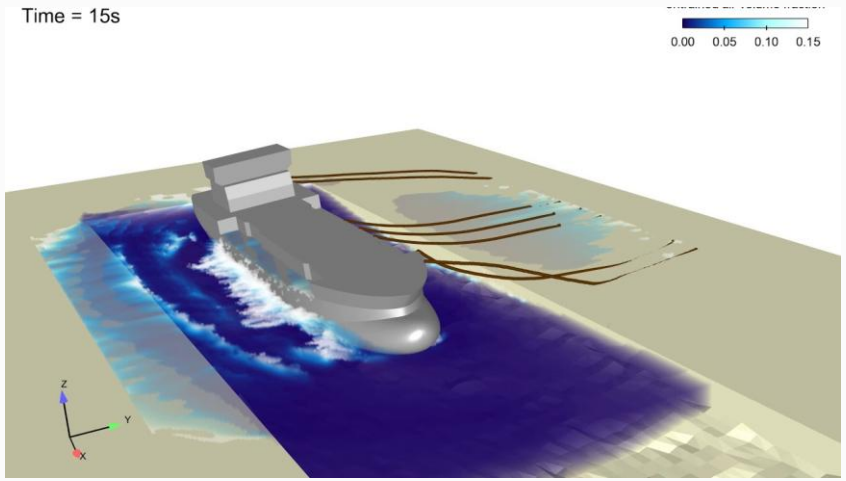

- Minimum energy states for a molecular configuration



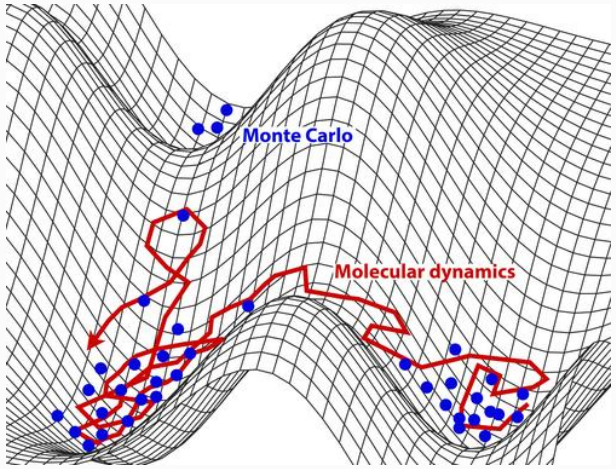

- – Plasma equilibrium configurations in a tokamak reactor - a perfect balance between pressure gradients and magnetic field

# Time-varying problems

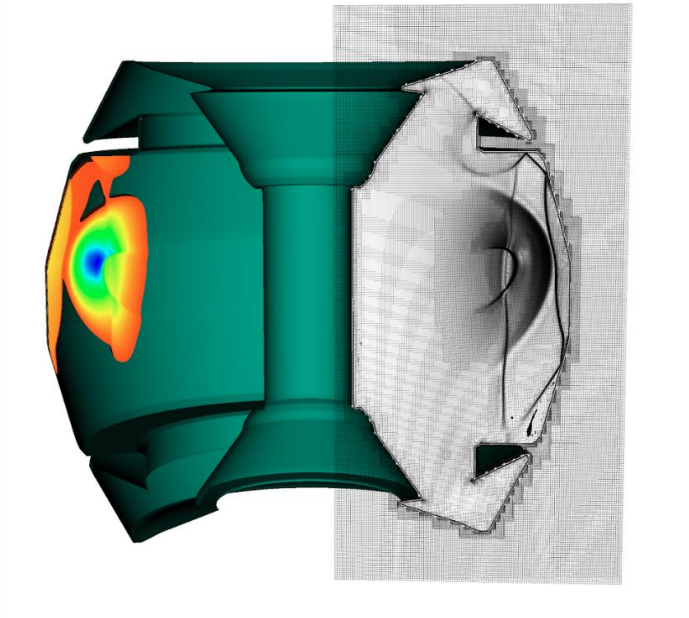
- Alternatively, our solution may continuously change in space and time
- Here, examples include:




- Water flow around an object as it moves through the real world




- Evolution of atoms according to molecular dynamics



- Plasma instabilities in a reactor – a perfect balance is not physically attainable

# Mathematical classification

- Whilst steady state and time-varying do describe features of the equations, this doesn't necessarily help us decide upon our numerical method
- Mathematically, we use the slope (derivative) of our variables to identify how things are changing, and things change according to some propagation of information
- We consider **speeds of information propagation**
    - Equations where **all** information propagates infinitely fast

      **Elliptic equations**
    - Equations where **some** information propagates infinitely fast, but some travels with a finite speed

      **Parabolic equations**
    - Equations where **all** information travels at a finite speed

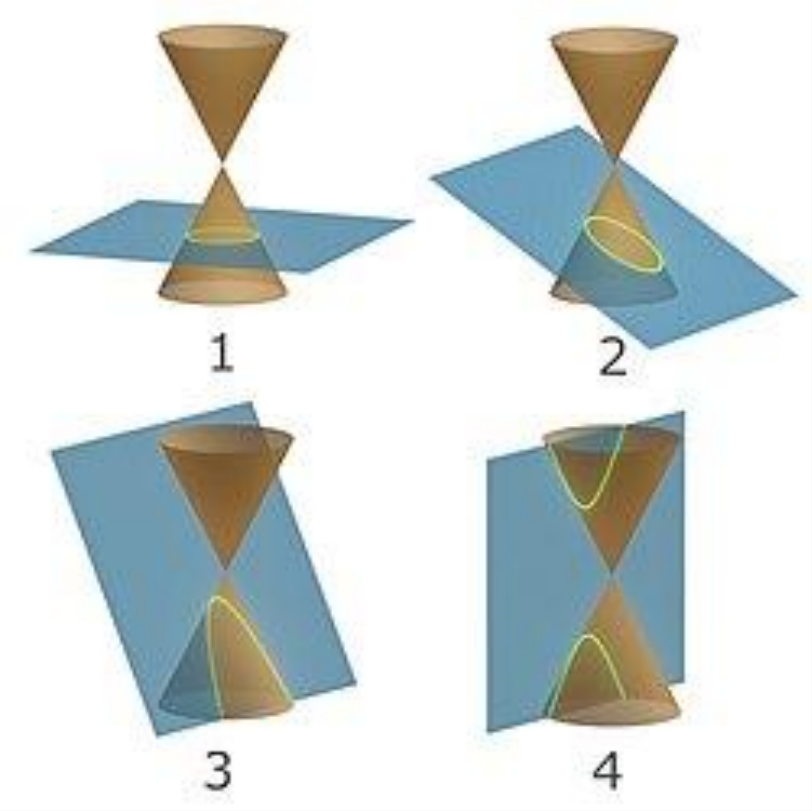
      **Hyperbolic equations**
- The behaviour in parabolic equations is often also referred to as **diffusion**

# Classification — conic sections

- Classification of PDEs is actually a wide area with ongoing research — we shall only consider a few examples
- One example starts with drawing comparisons to conic sections (the lines through which a cone can be cut)
- Such lines are parameterised through

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- The sign of the discriminant, $\mathcal{D} = B^2 - 4AC$, determines the shape:
  - $\mathcal{D} < 0$: Ellipse
  - $\mathcal{D} = 0$ : Parabola
  - $\mathcal{D} > 0$ : Hyperbola



# Classification — linear second-order PDEs

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- There is a similarity between conic sections and linear second-order PDEs

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + F = 0$$

- We can still consider the discriminant, $\mathcal{D} = B^2 - 4AC$, now we have
  - $\mathcal{D} < 0$: Elliptic equation
  - $\mathcal{D} = 0$ : Parabolic equation
  - $\mathcal{D} > 0$ : Hyperbolic equation
- However, this is a very limited set of equations — what happens if you are modelling something that is not two-dimensional, not linear, not second order and/or not a single variable?

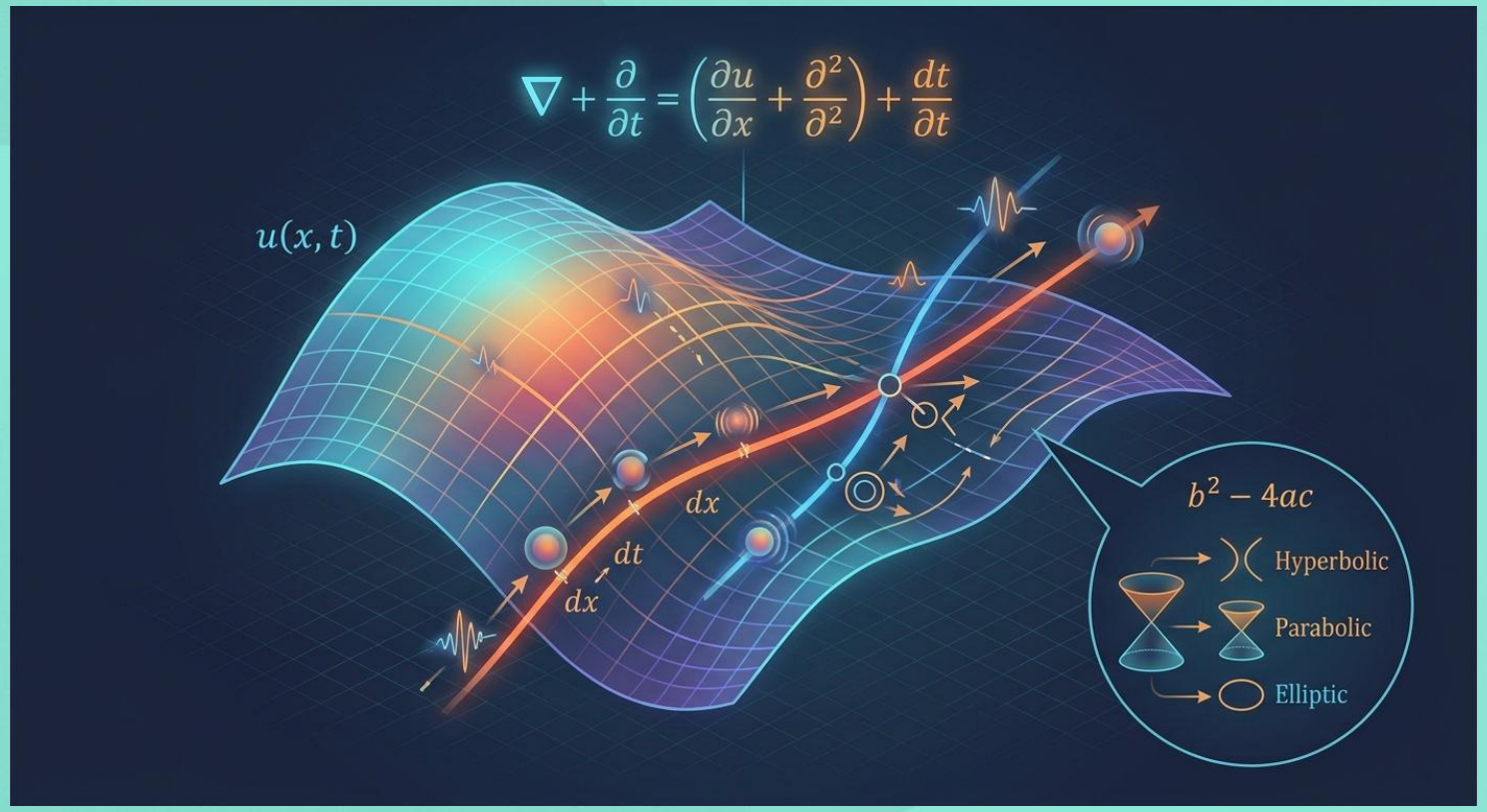
# Classification of the Laplacian

# Classification — systems

- Full classification of PDEs involves analysis of **characteristics** (which govern information propagation) — this can get complex very quickly
- However, sometimes equations can be written in the form: $$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0$$
- With a bit of multivariable chain rule, this can be expressed as $$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \nabla \cdot \mathbf{u} = 0$$
- The eigenvalues of the Jacobian matrix $\partial \mathbf{f} / \partial \mathbf{u}$ **might** allow us to classify the system of PDEs:
  - All purely complex: Elliptic system
  - All real and distinct: Hyperbolic system
- Other classifications are not so straightforward, and need more investigation – nonlinear equations could sometimes have conditionally real eigenvalues, i.e. some physical situations could provide imaginary values

# Classification of simple PDEs

# *Different Forms for PDEs*



# Different forms of PDEs

- Throughout the MPhil in Scientific Computing, you will be looking at solutions to various partial differential equations
- In general, these can all be written in a form which looks like $$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{s}(\mathbf{u})$$
- They will not always look like this, though
- Especially since the term $\mathbf{s}(\mathbf{u})$ can hide anything that is not a time derivative or divergence of a quantity
- However, no matter whether you start from the Boltzmann equation, Navier–Stokes equations, Maxwell's equations or even Einstein's field equations, they can all be written in this form (in some cases, with considerable effort)
- The reason we split up equations like this is because each term contributes something specific to the solution, and we can investigate this behaviour in isolation

# Commonly encountered forms

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{s}(\mathbf{u})$$

- Identifying behaviour often requires manipulating the equations, which itself involves playing with derivatives and independent variables – it is important to know the various ways in which equations can be expressed
- For simplicity, we shall consider one dimension, and $\mathbf{s} = \mathbf{0}$, but the way we manipulate derivatives here will also apply to any in this $\mathbf{s}$ term
- So, starting from $\frac{\partial u_i}{\partial t} + \frac{\partial f_i(\mathbf{u})}{\partial x} = 0$
- We could get:

$$\frac{\partial u_i}{\partial t} + A_{ij}\frac{\partial u_j}{\partial x} = 0 \qquad \frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0 \qquad \frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij}\frac{\partial \mathcal{V}_j}{\partial x} = 0$$

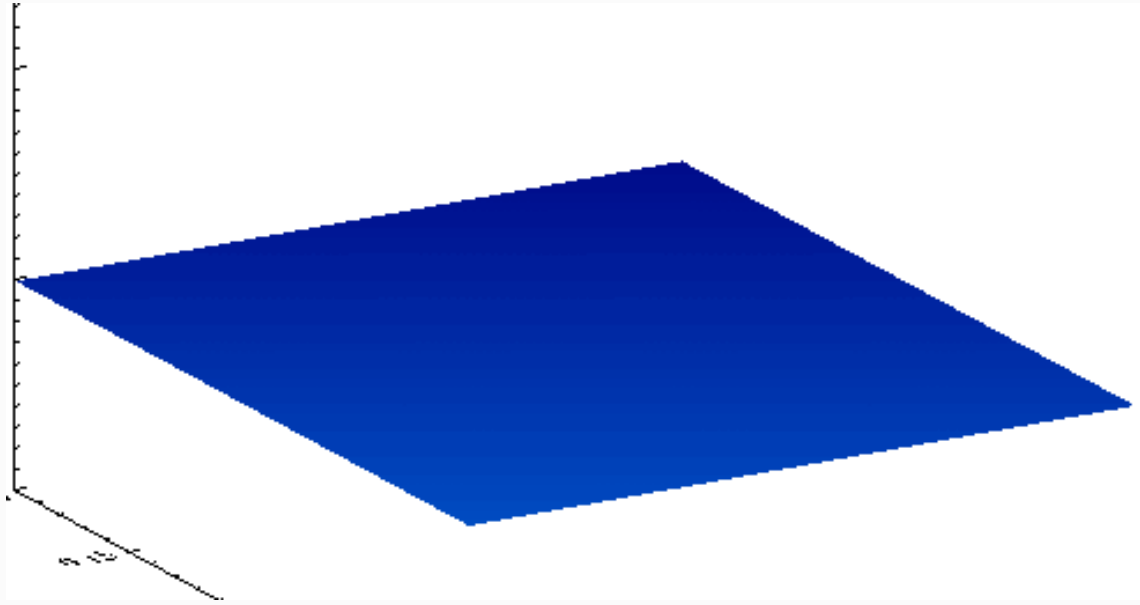
Where $\Lambda$ is a diagonal matrix

- These forms may allow for accurate numerical solution, analysis of the equations, or even identification of analytic solutions – we shall consider how, and why, we want these

# An example — the shallow water equations

- As a simple, but non-trivial, example to demonstrate these different forms, and some of the mathematical techniques used, we consider the shallow water equations
- They are a reduction of the Navier–Stokes equations, and are a suitable approximation for systems where there is a vertical length scale much smaller than other length scales (such as shallow water)

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2+\frac{1}{2}\phi^2\end{pmatrix}=\begin{pmatrix}0\\ g\phi\,\partial h/\partial x\end{pmatrix}$$

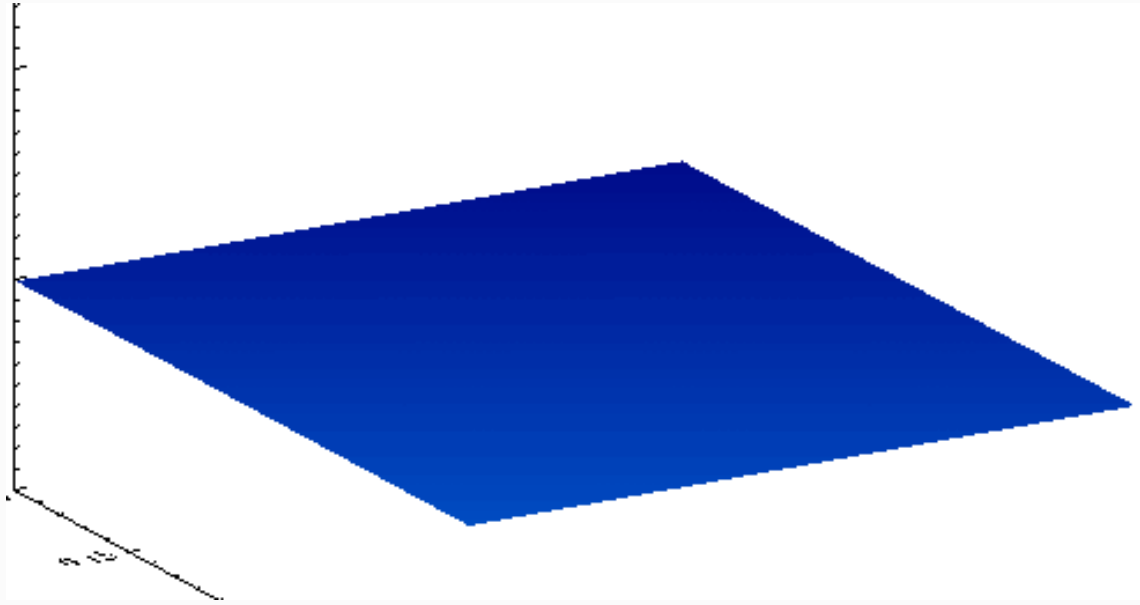


Source: Dan Copsey, own work, wikipedia

# An example — the shallow water equations

- In this one-dimensional form, $u$ is the velocity, $h$ the height (or depth) of the lower surface (e.g. the sea bed) and $g$ is gravitational acceleration
- $\phi$ is the geopotential, related to the height through $\phi = gH = g(\eta + h)$ where $\eta$ is the height above an initial surface
- If a uniform depth is considered, the right hand side vanishes (and since we are setting $\mathbf{s} = \mathbf{0}$, this is what we want)

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2 + \frac{1}{2}\phi^2\end{pmatrix} = \begin{pmatrix}0\\ g\phi\,\partial h/\partial x\end{pmatrix}$$



Source: Dan Copsey, own work, wikipedia

# Conservation form

$$\frac{\partial u_i}{\partial t} + \frac{\partial f_i(\mathbf{u})}{\partial x} = 0$$

- This form of equations is referred to as **conservation** form

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi \\ \phi u\end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix}\phi u \\ (\phi u)^2/\phi + \frac{1}{2}\phi^2\end{pmatrix} = \mathbf{0}$$

$\mathbf{u}$ is the **vector of conserved variables**, because this form of equations is a conservation law

- It is clear that the shallow water equations, as presented so far, are in this form

$$\mathbf{u} = \begin{pmatrix}\phi \\ \phi u\end{pmatrix}, \qquad \mathbf{f} = \begin{pmatrix}\phi u \\ (\phi u)^2/\phi + \frac{1}{2}\phi^2\end{pmatrix}$$

$\mathbf{f}$ is the **flux vector**, because it can describe the 'flux' or 'flow' of $\mathbf{u}$ through a volume

- Note that $\phi$ and $\phi u$ are **independent variables** in this case

# Conservation form

- The term conservation form is directly related to the fact that in multiple dimensions, our equations are

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0$$

Conservation form is a fundamental part of compressible fluid dynamics

- One definition of divergence of a vector field is that it measures the rate at which the vector field enters or leaves a volume

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$

- This is **conservation**; the only way to change $\mathbf{u}$ is by adding or removing some from a volume through its boundaries – if this doesn't happen, then the total amount of $\mathbf{u}$ doesn't change in the volume, even if its distribution does

- Alternatively, the divergence term could be balanced by a term on the right-hand side of the equation — we could have **balance law** form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{s}(\mathbf{u})$$

# Non-conservation form (and the Jacobian)

This form is perhaps the most common for a computational solution – we want to solve everything in terms of our independent variables $\mathbf{u}$

$$\frac{\partial u_i}{\partial t} + A_{ij}\frac{\partial u_j}{\partial x} = 0$$

- It can be good to have all derivatives being taken of the same vector, $\mathbf{u}$, this could be known as a non-conservative form (or just referred to as a PDE without further clarification)
- The matrix quantity $\mathrm{A}$ is the **Jacobian** of the vector $\mathbf{f}$

$$A_{ij} = \frac{\partial f_i}{\partial u_j} \qquad \text{and by the chain rule} \qquad \frac{\partial f_i}{\partial x} = \frac{\partial f_i}{\partial u_j}\frac{\partial u_j}{\partial x}$$

- For the shallow water equations, we get

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi \\ \phi u\end{pmatrix} + \begin{pmatrix}0 & 1 \\ -u^2+\phi & 2u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\phi \\ \phi u\end{pmatrix} = \mathbf{0}$$

- Because $\phi$ and $\phi u$ are **independent** variables, $\dfrac{\partial(\phi u)}{\partial \phi} = 0$

# Non-conservation form

# Primitive-variable form

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

- Sometimes the conserved variables are not the most intuitive variables to use, and the shallow water equations are a good example of this
- Why use geopotential if we could use height?
- Alternative variable choices are sometimes called **primitive variables** (if you have a reference set of variables), and may allow for more intuitive or analytic understanding of the equations
- To qualify as a set of primitive variables, we need variables which allow for a **closed system of equations**, but they do not form a unique set of variables – these could just be considered another form of the (non-conservative) PDE
- In the shallow water equations, a suitable choice of primitive variables might be total height, $H$, and velocity, $u$
- But we could also have $\phi$ and $u$, or $\eta$ and $u^2$, but **not** $\phi$ and $H$ (from which u cannot be computed)

# Primitive-variable form (continued)

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

Even if your system of equations doesn't have the common concept of primitive variables, we are really introducing this to show how variable transformations happen

- Obtaining primitive-variable form is now a matter of multivariable chain rule

$$\frac{\partial u_i}{\partial w_j}\frac{\partial w_j}{\partial t} + \frac{\partial f_j(\mathbf{w})}{\partial w_j}\frac{\partial w_j}{\partial x} = 0$$

- Therefore

$$B_{ij} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k(\mathbf{w})}{\partial w_j} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k}{\partial u_l}\frac{\partial u_l}{\partial w_j}$$

- This last expression implies $\mathrm{A}$ and $\mathrm{B}$ are **similar matrices**

# Primitive form

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# Primitive-variable form (continued)

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

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- Therefore
- This last expression implies A and B are **similar matrices**
- For the shallow water equations
- Often, it is not worth going through the matrix calculations, and it is much easier to do this by simple manipulation of the equations

- Therefore

$$B_{ij} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k(\mathbf{w})}{\partial w_j} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k}{\partial u_l}\frac{\partial u_l}{\partial w_j}$$

- This last expression implies $\mathrm{A}$ and $\mathrm{B}$ are **similar matrices**
- For the shallow water equations

$$\frac{\partial}{\partial t}\begin{pmatrix} H \\ u \end{pmatrix} + \begin{pmatrix} u & H \\ g & u \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} H \\ u \end{pmatrix} = \mathbf{0}$$

- Often, it is not worth going through the matrix calculations, and it is much easier to do this by simple manipulation of the equations

# Primitive form SW

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# Characteristic form

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij} \frac{\partial \mathcal{V}_j}{\partial x} = 0$$

- Characteristic form allows for a wide range of analysis of partial differential equations
- It can be used for finding exact solutions in some cases, approximate solutions in others, or understanding the behaviour of the system
- It is important to note that not all equations can be written in characteristic form; since $\Lambda$ is a diagonal matrix, at some point we will need to diagonalise a matrix to obtain it, which may not be possible
- We could consider characteristic form to be a very particular case of primitive variable form

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + B_{ij} \frac{\partial w_j(\mathbf{u})}{\partial x} = \frac{\partial w_i(\mathbf{u})}{\partial t} + (C_{ik})^{-1} A_{kl} C_{lj} \frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

# Characteristic form (continued)

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij}\frac{\partial \mathcal{V}_j}{\partial x} = 0$$

- We need to find a matrix, $C$, such that $(C_{ik})^{-1}A_{kl}C_{lj}$ is diagonal

$$\frac{\partial w_i(\mathbf{u})}{\partial t} + (C_{ik})^{-1}A_{kl}C_{lj}\frac{\partial w_j(\mathbf{u})}{\partial x} = 0$$

- In doing so, we will effectively choose the 'primitive' variables for this case, since

$$C_{ij} = \frac{\partial u_i}{\partial w_j} = \frac{\partial u_i}{\partial \mathcal{V}_j}$$

- This may well be familiar — we are **diagonalising the Jacobian** to achieve this
- Therefore the entries of $\Lambda$ are the eigenvalues of $\mathrm{A}$, and $C_{ik}$ and $(C_{ik})^{-1}$ are matrices of the eigenvectors and its inverse respectively

# Characteristic form SW

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# Shallow water equations' eigenvalues and eigenvectors

- Two eigenvalues,

$$\lambda_1 = u + \sqrt{\phi} = u + \sqrt{gH}, \qquad \lambda_2 = u - \sqrt{\phi} = u - \sqrt{gH}$$

- These eigenvalues are both **real and distinct** — we know we can write the equation in characteristic form as a result
- The right eigenvectors of $\mathrm{B}$ are

$$\mathbf{r}_1 = \begin{pmatrix} H/\sqrt{gH} \\ 1 \end{pmatrix}, \qquad \mathbf{r}_2 = \begin{pmatrix} -H/\sqrt{gH} \\ 1 \end{pmatrix}$$

- The matrix of right eigenvectors is then

$$\frac{\partial \mathbf{w}}{\partial \mathcal{V}} = \begin{pmatrix} H/\sqrt{gH} & -H/\sqrt{gH} \\ 1 & 1 \end{pmatrix}$$

# Shallow water equations' eigenvalues and eigenvectors

$$\frac{\partial \mathbf{w}}{\partial \mathcal{V}} = \begin{pmatrix} H/\sqrt{gH} & -H/\sqrt{gH} \\ 1 & 1 \end{pmatrix}$$

- The inverse of this matrix is now

$$\left(\frac{\partial \mathbf{w}}{\partial \mathcal{V}}\right)^{-1} = \frac{1}{2}\begin{pmatrix} \sqrt{gH}/H & 1 \\ -\sqrt{gH}/H & 1 \end{pmatrix}$$

- We now have, as expected,

$$\Lambda = \begin{pmatrix} u+\sqrt{gH} & 0 \\ 0 & u-\sqrt{gH} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \sqrt{gH}/H & 1 \\ -\sqrt{gH}/H & 1 \end{pmatrix}\begin{pmatrix} u & H \\ g & u \end{pmatrix}\begin{pmatrix} H/\sqrt{gH} & -H/\sqrt{gH} \\ 1 & 1 \end{pmatrix}$$

- Now all we need is $\boldsymbol{\mathcal{V}}$

# Characteristic form of the shallow water equations

- Note that (for well behaved $\mathbf{w}$ and $\boldsymbol{\mathcal{V}}$)

$$(C_{ij})^{-1} = \left(\frac{\partial w_i}{\partial \mathcal{V}_j}\right)^{-1} = \frac{\partial \mathcal{V}_i}{\partial w_j}$$

- This means we can integrate both sides with respect to $\mathbf{w}$ to get $\boldsymbol{\mathcal{V}}$

$$\boldsymbol{\mathcal{V}} = \begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix}$$

- This gives

$$\frac{\partial}{\partial t}\begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix} + \begin{pmatrix} u + \sqrt{gH} & 0 \\ 0 & u - \sqrt{gH} \end{pmatrix} \frac{\partial}{\partial x}\begin{pmatrix} \sqrt{gH} + \frac{1}{2}u \\ \sqrt{gH} - \frac{1}{2}u \end{pmatrix} = \mathbf{0}$$

Characteristic form is not pretty, but it is powerful, the quantities in the diagonal matrix tell us a lot about how the system behaves, and how the variables are coupled

# Summary

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2+\frac{1}{2}\phi^2\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\begin{pmatrix}0 & 1\\ -u^2+\phi & 2u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}H\\ u\end{pmatrix}+\begin{pmatrix}u & H\\ g & u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}H\\ u\end{pmatrix}=\mathbf{0}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\sqrt{gH}+\frac{1}{2}u\\ \sqrt{gH}-\frac{1}{2}u\end{pmatrix}+\begin{pmatrix}u+\sqrt{gH} & 0\\ 0 & u-\sqrt{gH}\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\sqrt{gH}+\frac{1}{2}u\\ \sqrt{gH}-\frac{1}{2}u\end{pmatrix}=\mathbf{0}$$

- Four different ways to express the same equation
- They are all used at different times, and are not an exhaustive list of how we could express this equation

# Second-order PDEs

- Another way equations can be manipulated is that second-order PDEs can be obtained from a system of first-order PDEs (or vice-versa)
- Consider

$$\frac{\partial}{\partial t}\begin{pmatrix}\bar{u}\\ \bar{v}\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\bar{v}\\ a^2\bar{u}\end{pmatrix}=0$$

- If we take additional derivatives of $\bar{u}$ with respect to time and of $\bar{v}$ with respect to space, we get

$$\frac{\partial^2\bar{u}}{\partial t^2}=a^2\frac{\partial^2\bar{u}}{\partial x^2}$$

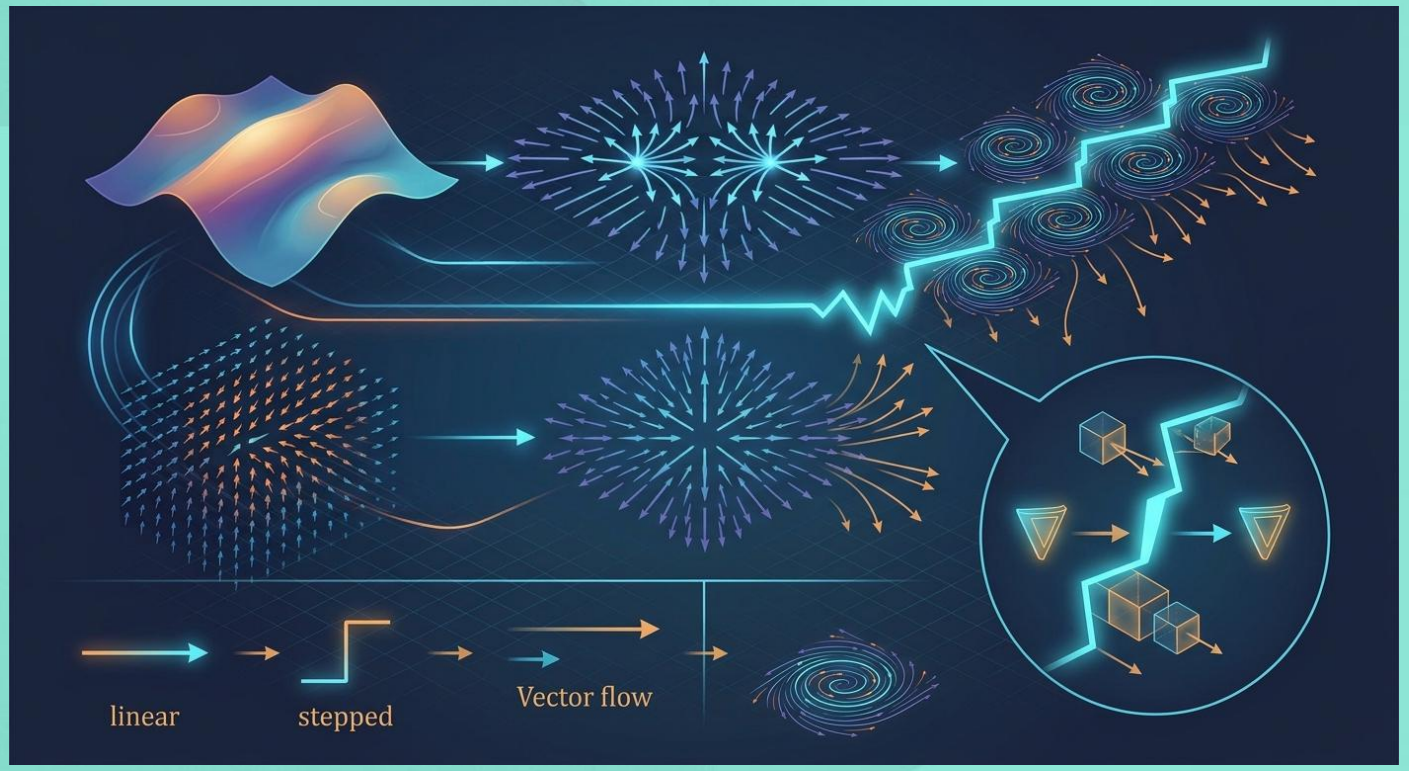
- This is the wave equation, and we know we can write it as a system of first-order PDEs (which may be easier to solve)
- However, working backwards is not always as straightforward — reintroducing $\bar{v}$ by integrating the wave equation is difficult

# Wave equation

# Wave equation

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# *Discontinuities and PDEs*



# Integral form of a differential equation

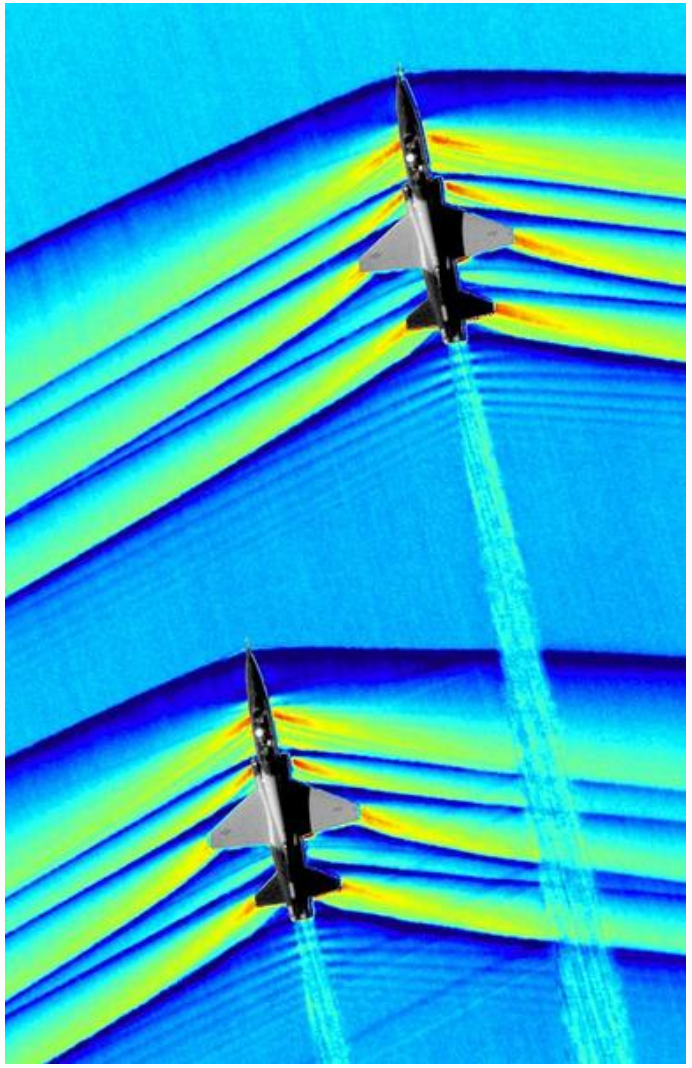
- It is probably most common to find the equations in scientific computing written as (partial) differential equations, e.g.

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{f}(u) = 0$$

- There are sometimes reasons that we don't want to do this, though
- For example, for this equation to be valid, we require that $u$ is differentiable
- Mathematically, there is no reason for any arbitrary function $u(\mathbf{x}, t)$ that this is true and it is possible for functions which are initially smooth to evolve to be non-differentialble
- If this happens, we cannot solve this form of the equations, but must instead solve the **integral form**
- In other cases, numerical methods may actually work better if everything is written as an integral – you may be able to make use of orthogonality properties or make your boundary conditions easier

# Discontinuities

- When we talk about a solution becoming non-differentiable, we are also saying that our function $u$ becomes **discontinuous**
- Common types of discontinuity are material interfaces and shock waves
- In reality, neither of these is a true discontinuity; there is always some form of transition, which, if you are on the atomistic stream, you would be resolving
- Thermal diffusion, no matter how small, exists over shock waves
- Materials undergo a transition region, which may be a region on the scale of just a few atoms, but noticeable all the same
- But there are good reasons that our equations allow discontinuities to form



Source: nasa.gov

# Singularities

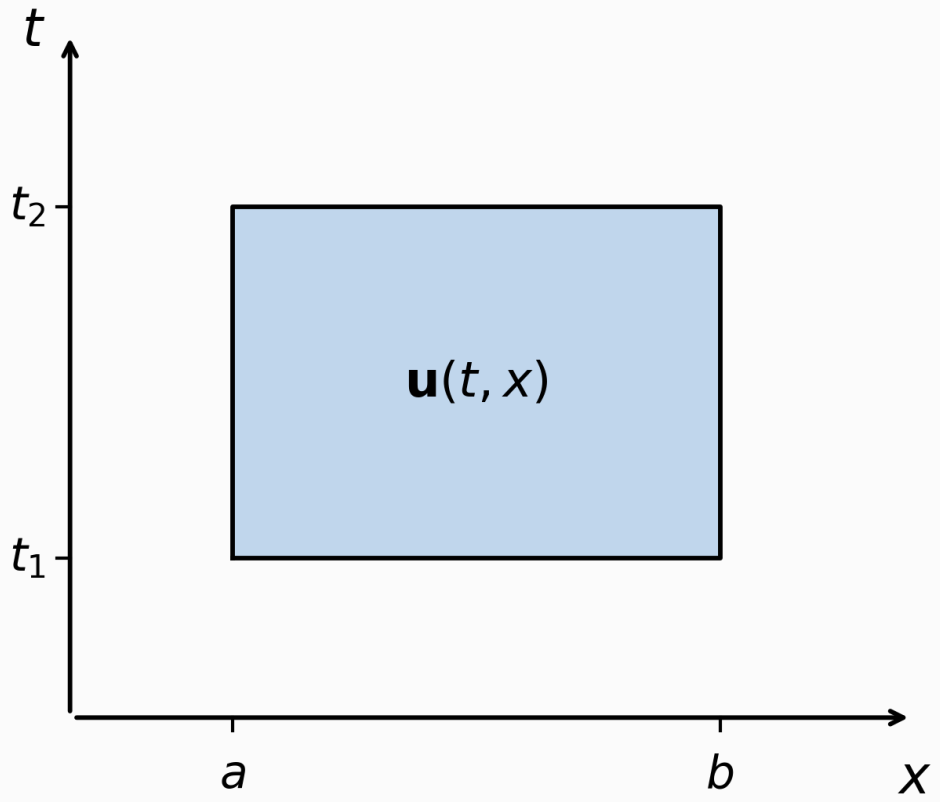
- Referring to an equation as forming a singularity is another way in which this mathematical behaviour is often described
- From a continuum point of view, features like shock waves and material interfaces are often best treated as true discontinuities because the simulation cannot model the true transition scales
- Elsewhere, singular behaviour occurs when your equations stop being physically valid – either by design (you need to know when the black hole forms) or by limitation of a model (you are moving into a new physics/chemistry regime)
- This means sometimes you stop your simulation at this point, but integral form exists for when you need to continue!



Source: https://www.eso.org/public/images/eso0903a/

# Integral domain

- Consider a differential equation $$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$
- In order to convert this to an integral equation, we first define some area we are trying to solve this equation over
- Typically, we would consider the case in which we know the solution at a point in time, $u(t_1, x), \quad x \in [a, b]$
- And we want to solve it for some (or all) times up until $t_2$, still for $x \in [a, b]$
- We can integrate our differential equation over this entire region



# Integral form

# Integral form and conservation laws

- Begin with the assumption that u is continuous, and integrate

$$\int_a^b \int_{t_1}^{t_2} \frac{\partial u(t,x)}{\partial t}\, dt\, dx + \int_{t_1}^{t_2} \int_a^b \frac{\partial f(u(t,x))}{\partial x}\, dx\, dt = 0$$

- We can then use the fundamental theorem of calculus

$$\int_a^b u(t_2,x) - u(t_1,x)\, dx + \int_{t_1}^{t_2} f(u(t,b)) - f(u(t,a))\, dt = 0$$

- In this form, it would not matter if u was discontinuous — a solution could still be obtained
- We shall not detail how you might now find a solution, for that you will need other courses
- For example, I will tell you how to do this for compressible gases in "*Numerical Methods for Compressible Fluid Dynamics*"

# Weak solutions

- A weak solution is defined as a solution of

$$\int_a^b u(t_2, x) - u(t_1, x)\, dx + \int_{t_1}^{t_2} f(u(t, b)) - f(u(t, a))\, dt = 0$$

  which may, or may not, be discontinuous

- If it is discontinuous, it cannot satisfy

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

- However, this discontinuous solution is indeed a (weak) solution to the underlying physical problem, just not one that can be found by considering this form of the governing equation (because this governing equation made assumptions about the solution which don't actually hold)

- It is important to note that when solving for $u$ on $x \in [a, b]$, $t \in [t_1, t_2]$, weak solutions may not be unique. If you stick with my lectures, you will see this, but otherwise you might imagine that this arises because different functions can have the same area (same integral) between $a$ and $b$, but would have different slopes